



DERIVATIVES – ADM 15535

Spring 2020

Course overview

This course is designed to be thorough introduction to derivatives products and markets. We will focus on describing the mechanics, pricing, hedging and applications of the basic linear (forwards, futures and swaps) and non-linear derivatives (options). This class will emphasize the relationship between non arbitrage and risk neutral valuation as the main cornerstone for derivative pricing. On a practical note, we will focus on the use of these instruments to transfer, mitigate, and eliminate different types on financial risk arising in industry situations. We will start by introducing the primary concepts in an intuitive manner and then, after formalizing these concepts in a more rigorous theoretical framework, we will follow with more complex examples and practical applications. We will also emphasize economic intuition, industry relevance and corporate applications whenever possible, especially in the second part of the course.

Learning objectives

By the end of the class, you should at least have a basic understanding of the workings and pricing of the main derivatives instruments and understand their usefulness and applications in a variety of industry settings. In addition, the class should provide a solid theoretical background for those who wish to take more advanced courses in derivatives, asset pricing, or other related topics.

More specifically, you should be able to:

- Describe the mechanics, payouts, and applications of the most common derivatives instruments: forwards, futures, swaps and options in its many variations and forms.
- Apply non arbitrage arguments and risk neutral valuation to price these derivatives instruments across asset classes (equity, currencies, commodities, interest rate products, etc.).
- Develop some familiarity with numerical methods, primarily binomial trees and Montecarlo simulation, that can be used to value more complex derivatives for which an analytical solution might not exists.
- Recognize and understand the main corporate applications of derivatives products, such as real options, warrants and convertible debt issuance.

Instructor

Jeffrey Sadowsky, email: Jeffrey.sadowsky@itam.mx.

Mr. Sadowsky, holds a PhD. in Management Science and Engineering and a MS in Financial Mathematics from Stanford University. He has more than 14 years of industry experience in fixed income derivatives markets and structured financing, having worked in JP Morgan, Credit Suisse, both in the New York and London offices, as well as Commerzbank AG. Currently, he is the Director of Treasury, Risk Management and Capital Markets at Konfio, a Mexican fintech lender.

Class Hours

Tuesday 4pm to 7pm (Session 1)

Tuesday 7pm to 10pm (Session 2)

Please be on time so as not to disturb the normal flow of the class and your other classmates. I don't want to impose a lot of specific rules, but I will ask you to always behave in a manner that is respectful, courteous, and mindful of your other classmates and your instructor.

Examinations and Grading

- The mid-term exam will be a 2.5-3 hour in-class exam, most likely a combination of a multiple choice section and a written section. You will be allowed to bring any summary sheet, class notes or academic reference. Any computing equipment can be used.
- The final exam will be a 3 hour exam on a date designated by the University. It will be on a similar format to the midterm (open book, part multiple choice, part written section).
- In addition, I will assign homework after each class. The homework won't be graded but it is a critical part of the learning process as it serves a twofold purpose: it is meant to consolidate your understanding of the concepts imparted in class via additional practical exercises, and it will help you prepare for the midterm and final exams. The homework questions have a similar style to those on the exam, so I would strongly encourage you to solve all the exercises assigned in due course.
- Between class attendance, readings and homework assignments, I expect you to devote at least 10-12 hours a week to learn the class material.

The grading will be based on the following formula:

max (40% midterm+60% final, 50% midterm+50% final)

Language requirements

The class will be given in Spanish. However, all bibliographical references, teaching materials and homework will be in English. At the very least, a **strong reading command of the English language** will be required.

Bibliographic References:

- i) **Robert L McDonald, *Derivatives Markets, International Edition (MD)*, Pearson:** this will be our main textbook and whose teaching materials we will use in class. However, this text will not be the main reference for every single topic, so please be prepared to in addition use the below references at some points in the course.
- ii) **John Hull: *Fundamentals of Futures and Options Markets, Fourth Edition, Prentice Hall (JH)*:** This text will be our complementary reference throughout the class and it is important that you are familiar with it. In addition, Hull's development of the topics is less idiosyncratic and his more advanced book *Option, Futures and Derivatives* is the main industry reference for the subject.
- iii) **David Luenberger, *Investment Science, Oxford University Press, First Edition (DL)*:** This is an excellent introduction to asset pricing, and a must-read for those who would like to pursue more advanced degrees. Luenberger is masterful at introducing technical concepts in a concise, clear and intuitive manner.

Detailed Content:

The topics below correspond are designed to be covered roughly in one session each. However, depending on the progress in class and student's feedback, more or less time than a full session might be allocated to each topic. **Please read the related chapters before class, at least from one if not two of the bibliographic references.**

- 1- Introduction: Definitions. Main types of Derivatives: Forwards, Futures and Options. Exchanges and OTC markets. Hedgers vs Speculators. The No Arbitrage principle. Margin trading and short-selling. (MCD Ch 1 and 2, H Ch 1 and 2)
- 2- Forwards and Futures: Forward pricing using the non-arbitrage principle. Currency Forwards. Commodity Forwards. Lease Rate and Convenience Yield. Futures and its difference with Forwards. (MCD Ch 8 and 7, JH Ch 3, DL Ch 10)
- 3- Introduction to Risk Management: Hedging with Forwards and Futures. Pros and Cons of Hedging. Basis Risk. Minimum Variance Hedge. Options as Insurance. (MCD Ch 5, JH Ch 4)
- 4- Introduction to Fixed Income: Bond Basics. Spot/Zero and Forward Rates. Forward Rate Agreements. Eurodollar and Treasury Bond Futures. Duration and Convexity (MCD Ch 9, JH Ch 5, DL Ch 3 and 4)
- 5- Swaps: Interest Rate Swaps (IRS). Mechanics and uses. Valuation of Interest Rate Swaps. Currency Swaps. Total Return Swaps (MCD Ch 10, JH Ch 6, DL Ch 10).
- 6- Introduction to Options: Types of Options. Payouts and mechanics. Put Call Parity. Upper and Lower Bound on Option Prices. Basic option strategies (spreads, collars, straddles, strangles). Applications to Risk Management. (MCD Ch 12, JH Ch 7.1-7.3, 8 and 9)
- 7- Introduction to Option Pricing: Basic replication argument. Two period binomial tree. Risk Neutral Probability. Multiperiod extensions and application to puts and American options (MCD Ch 13, JH Ch 10, DL Ch 12)
- 8- The Black Scholes Formula: Binomial Tree and Lognormality. Modeling Stock Returns. Basic Derivation of the Black Scholes Formula. Application to options on dividend paying assets and currency options. (MCD Ch 15 and Ch 23, JH Ch 11 and 12, DL Ch 11)
- 9- Black Scholes Greeks: Delta, gamma, theta, vega and rho. Market making and delta hedging. Option Elasticity. (MCD 15 and 16, JH Ch 15, DL Ch 13)
- 10- Implied Volatility: Volatility smile and volatility skew. Volatility Modeling and Estimation, ARCH and GARCH (MCD Ch 29, JH Ch 14)
- 11- Montecarlo Simulation: Simulating random walks and log normal stock prices. Montecarlo valuation of European and path dependent options. Introduction to Jumps. Estimation errors and Variance Reduction Techniques (MCD Ch 24)
- 12- Asset Pricing Theory: Non Arbitrage Revisited. Non Arbitrage, the Portfolio Choice Problem and Risk Neutral Valuation. Martingale Pricing. Revisiting the Black Scholes Formula using Risk Neutral Valuation (DL Ch 9 and 13, MCD 14.2 and 27)
- 13- Corporate Applications Part I: Introduction to Real Options. Investment and NPV Rule. Investing as an Option. Practical Applications:. Investment or abandonment, R&D as an Option, Commodity Extraction. (MCD Ch 21)
- 14- Corporate Applications Part II: Debt and Equity as Options. Debt, credit risk, expected return and yield to maturity. Warrants and Convertibles Bonds. Stock Options. Modeling acquisitions with options. (MCD Ch 20)

15- Introduction to Credit Derivatives: Merton's structural model. Default Probabilities, Exposure and Recovery. Survival, probabilities and hazard rates. Credit default swaps (CDS), mechanics and pricing. (MCD 27)